LocalMind

A Local-First Agentic Knowledge Copilot with a Custom MCP Memory Server



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Problem statement and objectives

Scattered files make useful project context hard to recover

Find local evidence

Search supported files in folders that the user approves.

Keep project context

Save explicit memories and retrieve them in later conversations.

Make answers inspectable

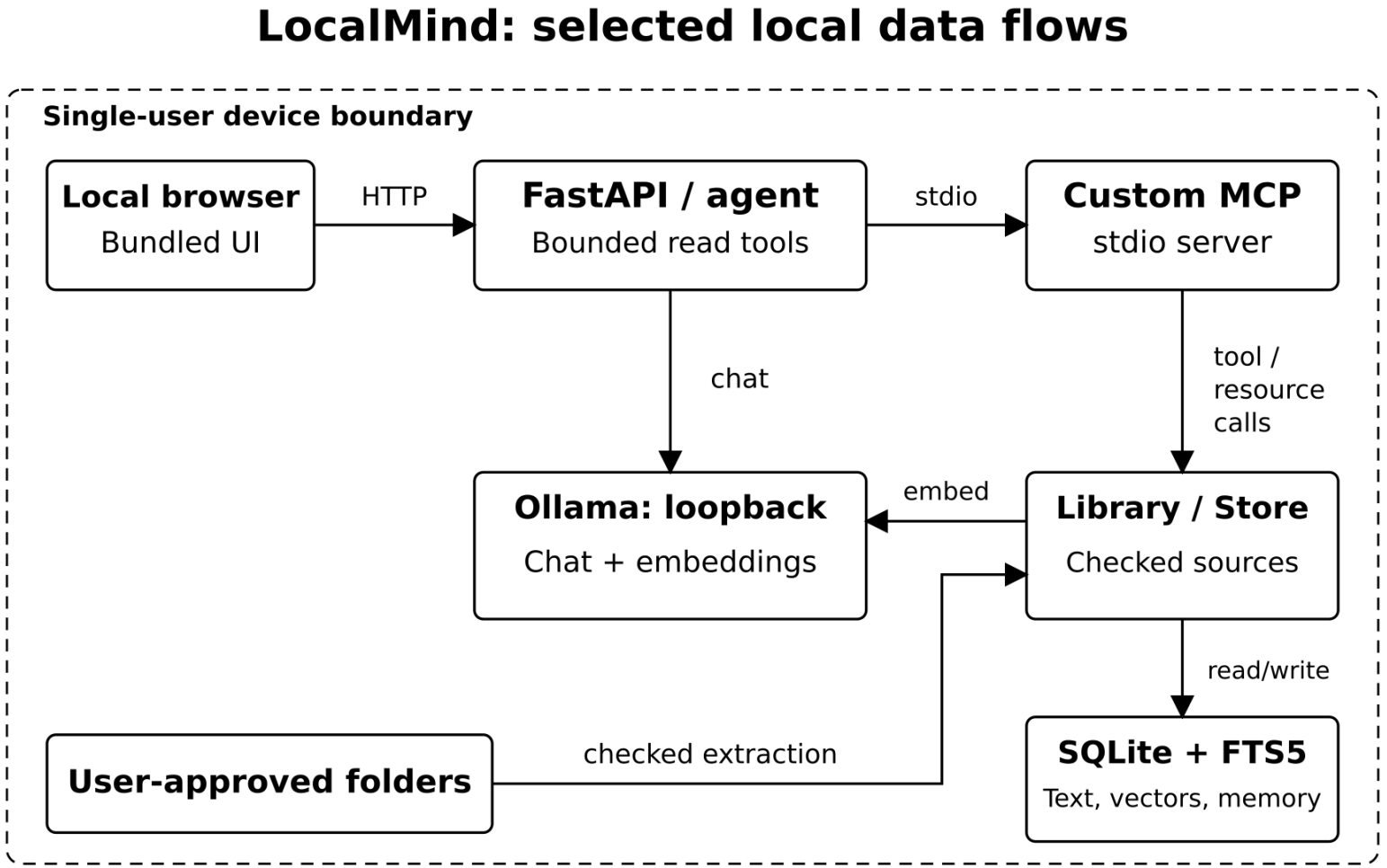
Show source documents and exact file identity beside responses.

Literature review and project positioning

RAG, Lewis et al. (2020) [1]	Retrieve evidence before generation
RRF, Cormack et al. (2009) [2]	Combine lexical and vector rankings
ReAct, Yao et al. (2023) [3]	Gather evidence through tool calls
MemGPT, Packer et al. (2023) [4]	Persist context beyond one model input
MCP specification (2025) [5]	Expose tools through a standard protocol

Contribution: a working local integration with reproducible evidence

System architecture



Approved folders

Extracted text and chunks enter the local index.

Local inference

Ollama serves chat and embedding models over loopback.

Retrieval, memory and MCP

Hybrid document retrieval

SQLite FTS5 finds words. EmbeddingGemma supplies vector similarity. RRF combines the rankings.

Persistent memory

SQLite stores conversations, project records and explicit saved memories across sessions.

Bounded agent tools

Seven read tools support search, document access, memory recall and catalog inspection.

Grounding controls

Exact file questions use the catalog

File lists and filename checks read index metadata through MCP.

Greetings avoid document retrieval

A neutral greeting needs no model call or project assumption.

Document answers retain inspectable evidence

Evidence records fit the serialized budget.
Failed reference checks return exact source excerpts.

A valid citation can still fail to support a claim

Implementation and laptop configuration

RTX 5050 Laptop GPU
Ryzen 7 250
16 GB RAM

Local application

FastAPI and Uvicorn
SQLite FTS5 and NumPy
MCP Python SDK and Ollama

Source default

Qwen3 4B
8,192 context tokens

Optional development profile

Qwen3.5 9B Q4_K_M
12,288 tokens, partial CPU offload

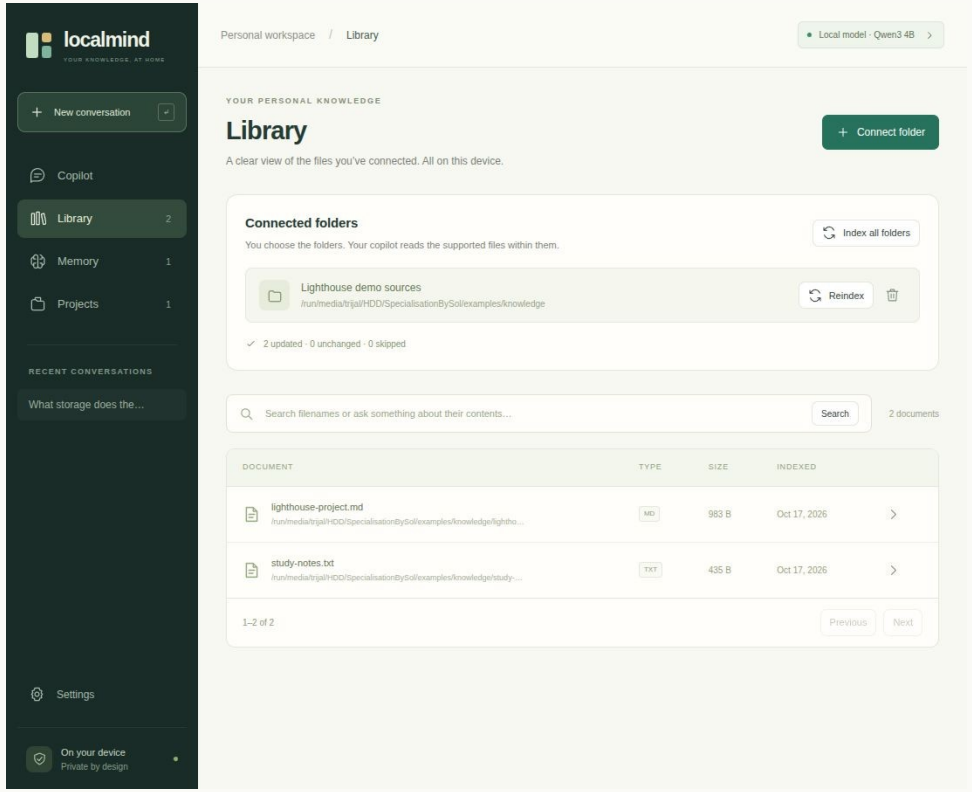
The frozen research benchmark used the earlier 4B profile

Two-minute demonstration

1. Open a source in Library
2. Ask a documented question
3. Compare the answer with its source
4. Save an explicit project memory

Question

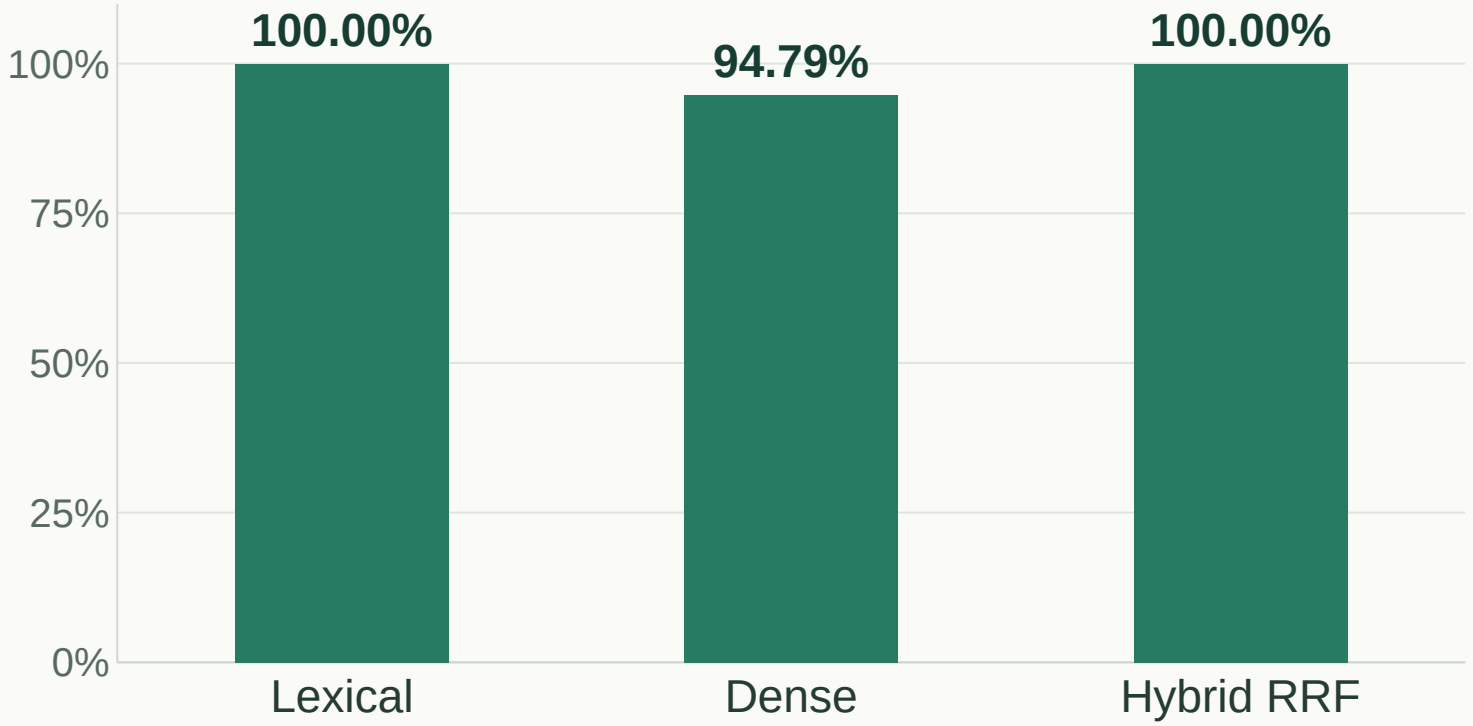
What storage does the Lighthouse project use?



Prepared sample workspace, 17 October 2026

Retrieval results on LM-Synth v1

39 synthetic documents, 141 chunks, 96 answerable questions



96/96

Lexical and hybrid cover every requested fact within five passages.

91/96 for dense retrieval

Evidence coverage on authored synthetic questions. This does not establish answer accuracy.

Generation results and the context repair

Complete answer matches on 24 reused synthetic questions

Condition	All-fact matches	Absence phrase
Original RAG and MCP	0/24 for each	8/8 for each
Repaired RAG	7/24 (29.17%)	8/8
Repaired MCP agent	3/24 (12.50%)	7/8

64/64 repaired requests retained initial evidence within budget

No additional model-selected tool calls occurred. The replay reuses inspected questions and is not held-out evaluation.

Verification and remaining limits

Software checks

163 Linux tests passed
200 Windows-copy tests passed on Linux
6 native checks skipped in that copy

Later native Windows CI: 206 tests
on each of Python 3.12 and 3.13

CI used a fake model service

Next evaluation and development

Held-out factuality and citation support

Stronger long-document coverage

Local OCR for scanned PDFs

Windows 11 laptop GPU measurements

SQLite currently has no
application-level encryption

Conclusion

A local knowledge workspace with persistent memory and inspectable sources

Approved local files supply evidence through retrieval and MCP.
Explicit memories preserve project context across sessions.

Measured retrieval coverage is strong on LM-Synth.
Broader answer reliability remains an evaluation priority.

Thank you to Dr. Sonali Kothari and the evaluation panel

Selected references

[1] P. Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks.” NeurIPS, 2020. arXiv:2005.11401.

[2] G. V. Cormack et al. “Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods.” SIGIR, 2009. DOI: 10.1145/1571941.1572114.

[3] S. Yao et al. “ReAct: Synergizing Reasoning and Acting in Language Models.” ICLR, 2023. arXiv:2210.03629.

[4] C. Packer et al. “MemGPT: Towards LLMs as Operating Systems.” arXiv:2310.08560, 2023, revised 2024.

[5] Model Context Protocol contributors. Model Context Protocol Specification, revision 2025-11-25.

[6] N. F. Liu et al. “Lost in the Middle: How Language Models Use Long Contexts.” TACL, 2024. DOI: 10.1162/tacl_a_00638.